Catholic Junior College

JC2 Preliminary Examinations

Higher 1

CANDIDATE NAME	
CLASS	**2T**

CHEMISTRY

Structured Questions

8873/02
17 August 2018
2 hours

Candidates answer on the Question Paper.
Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Section A – Answer **all** the questions**.**
Section B – Answer **one** question.

The use of an approved scientific calculator is expected where appropriate.
You are reminded of the need for good English and clear presentation in your answers.

A Data Booklet is provided.

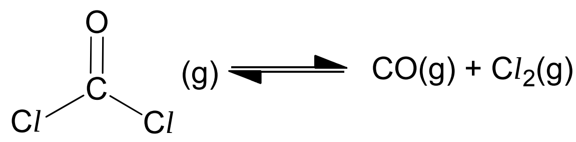
The number of marks is given in brackets [ ] at the end of each question or part of the question.

Section A	**Q1**	**12**
	Q2	**6**
	Q3	**15**
	Q4	**6**
	Q5	**6**
	Q6	**15**
Section B	**Q7**	**20**
	Q8	**20**
	Paper 2	**80**
	Paper 1	**30**
	Percentage	
	Grade	

Section A

Answer **all** the questions in this section in the spaces provided.

1 (a) Phosgene, $COCl_2$ is a gas with many industrial uses. At high temperature, gaseous phosgene decomposes to carbon dioxide and chlorine in a dynamic equilibrium according to the following equation:



phosgene

(i) Explain what is meant by the term *dynamic equilibrium.*

At dynamic equilibrium, the **rates of the forward and reverse reactions are equal** while there is **no change** in concentration/no. of mole of reactants and products.

(ii) Write an expression for the equilibrium constant, K_c, for the reaction.

Calculate its value when, at equilibrium, the concentrations of $COCl_2$(g), CO(g), and Cl_2(g) are 0.219 mol dm^{-3}, 0.530 mol dm^{-3} and 0.530 mol dm^{-3} respectively.

$$K_c = \frac{[CO][Cl_2]}{[COCl_2]}$$

$$K_c = \frac{(0.530)(0.530)}{(0.219)} = 1.28 \text{ mol dm}^{-3}$$

[2]

(iii) A certain amount of phosgene was added and the new equilibrium concentration of phosgene is 0.290 mol dm^{-3}. Calculate the new equilibrium concentration of CO(g).

Concentration of CO(g) and Cl_2(g) are equal as seen in part (ii)

$$K_c = \frac{[CO]^2}{[COCl_2]} = \frac{[CO]^2}{0.290} = 1.28 \text{ mol dm}^{-3}$$

new equilibrium concentration of CO(g) = 0.609 mol dm^{-3}

(iv) Calculate the amount in moles of phosgene added in part **(iii)**, given that the volume of the container is 3 dm^3.

	$COCl_2(g)$	$\rightleftharpoons$	$CO(g)$ +	$Cl_2(g)$
Initial amount / mol	3(0.219) + n = 0.657 + n		3(0.530) = 1.59	3(0.530) = 1.59
Change / mol	– 0.237		+0.237	+0.237
Equilibrium amount / mol	3(0.290) = 0.870		3(0.609) = 1.827	3(0.609) = 1.827

[The ICE table is not necessary. It helps to visualise the changes]

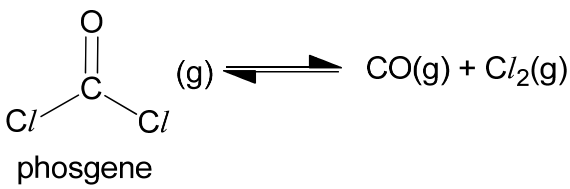
Let the amount added be n.
0.657 + n – 0.237 = 0.870
n = 0.450 mol

[2]

(b) Predict and explain how a decrease in pressure of the system will affect the position of equilibrium.

When the pressure of the system is decreased, the system will react to increase the pressure by increasing the number of moles of gas , thus the position of equilibrium will shift to the right. [2]

(c) **(i)** By using bond energy values from the *Data Booklet,* calculate the enthalpy change for the decomposition of phosgene as shown below:



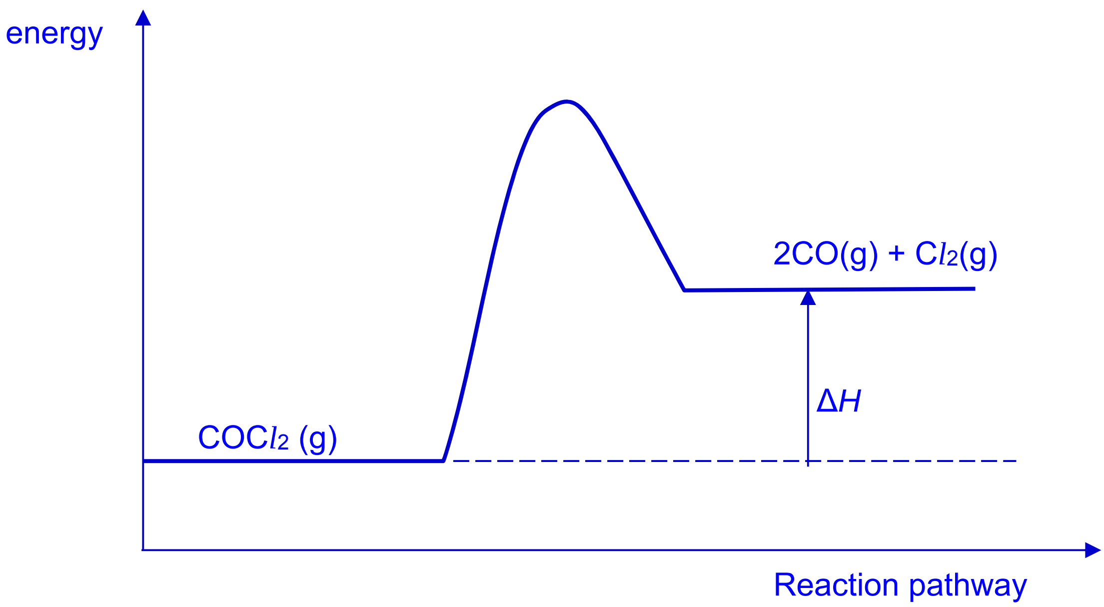
Bonds broken		Bonds formed	
1 C=O	**740**	1 C≡O	1077
2 C-C*l*	2(340)	1 C*l*-C*l*	244

ΔH = (740 + 2 × 340) – (1077 + 244)

= 1420 – 1321 = + 99.0 kJ mol^{-1}

[2]

(ii) Use your answer from part **(i)** to sketch a reaction pathway diagram for the decomposition of phosgene. Draw a labelled arrow to show the enthalpy change on your diagram.

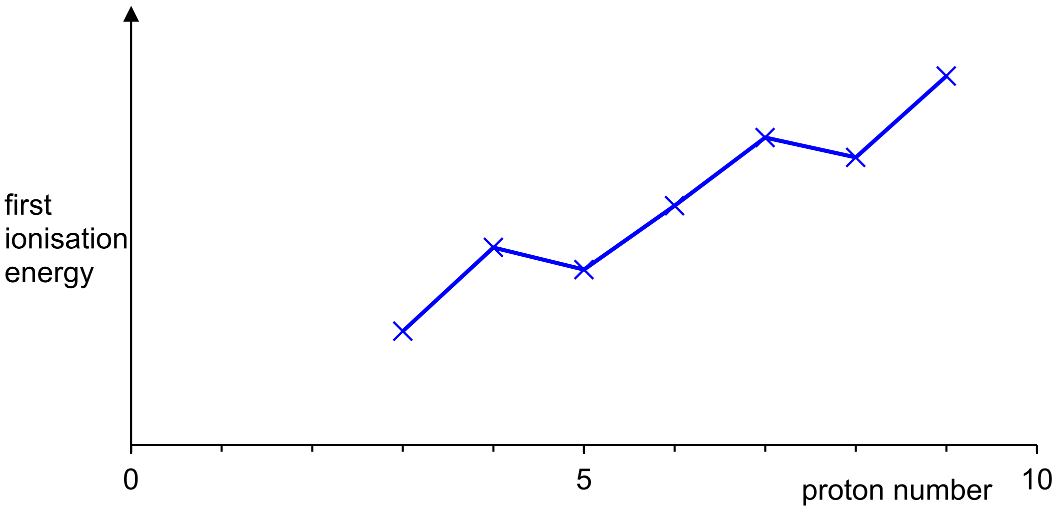


[2]
[Total: 12]

2(a) Write an equation to show the first ionisation energy of oxygen.

$O(g) \rightarrow O^{+}(g) + e^{-}$ **[with state symbols]**

(b) Sketch how the first ionisation energies of the elements change from lithium to fluorine.



(c) Explain why the first ionisation energy of oxygen is lower than

- nitrogen

N $1s^2\ 2s^2\ 2p_x^1\ 2p_y^1\ 2p_z^1$

O $1s^2\ 2s^2\ 2p_x^2\ 2p_y^1\ 2p_z^1$

Oxygen has a lower first ionisation energy than nitrogen, as less energy is required to remove an electron from paired $2p_x$ electrons in oxygen since **inter-electron repulsion** is experienced between the paired electrons.

[2]

- fluorine

The valence electrons of both oxygen and fluorine atoms experience **similar shielding effect,** but oxygen has **and larger size**, and **smaller nuclear charge**

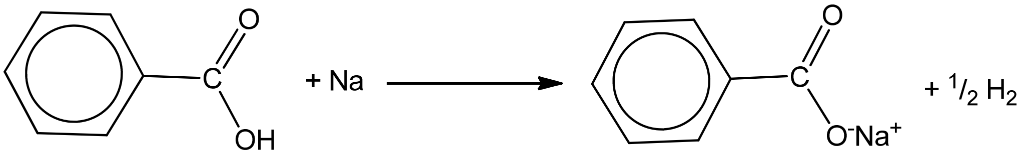
The **attraction** between outermost electrons and nucleus **is lower for oxygen** , resulting in less energy required to remove the outermost electron.

[2]

[Total: 6]

3 Benzoic acid, $C_6H_5CO_2H$, and its salts are often used as food preservatives in fruit juices and other acidic food. They are represented by the E-numbers, E210 to E213 when added as preservatives since they are able to inhibit the growth of mould, yeast and some forms of bacteria.

(a) Sodium metal is added to form the salt of benzoic acid, as shown in the following equation.



benzoic acid sodium benzoate

(i) 5.00 g of sodium metal was added to 0.01 moles of benzoic acid. Calculate the volume of gas produced at standard temperature and pressure for the above reaction.

No. of moles of Na = $\frac{5.00}{23.0}$ = 0.2174

No. of moles of benzoic acid given as 0.01 moles (limiting reagent)

No. of moles of gas formed = 0.01 x 0.5 x **22.7**
=0.114 dm^3 (to 3 s.f)

[2]

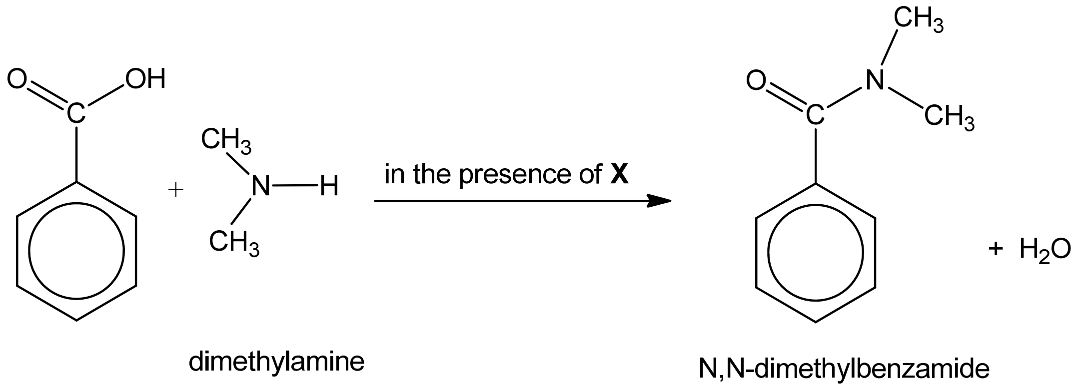
(ii) Explain why sodium benzoate is highly soluble in water but benzoic acid is only partially soluble in water.

Sodium benzoate is able to form ion-dipole interactions with water while the $-CO_2H$ group of the benzoic acid molecule is only able to form hydrogen bonds with water. The phenyl group of the benzoic acid molecule is non-polar and hence is only able to form instantaneous dipole-induced dipole interaction with water.

Hence, more energy is released in the formation of ion-dipole interaction between sodium benzoate and water which is sufficient to overcome the ionic bonds in sodium benzoate and hydrogen bonding between water.

[4]

(b) Benzoic acid can also react with amines to produce amides as shown in the following equation.



(i) Name the type of reaction shown by the above equation.

Condensation reaction ....

(ii) Identify **X** which is required for the above reaction to occur.

DCC ....

(iii) State the shape and bond angle around the N atom in dimethylamine.

Shape: ....Trigonal pyramidal..

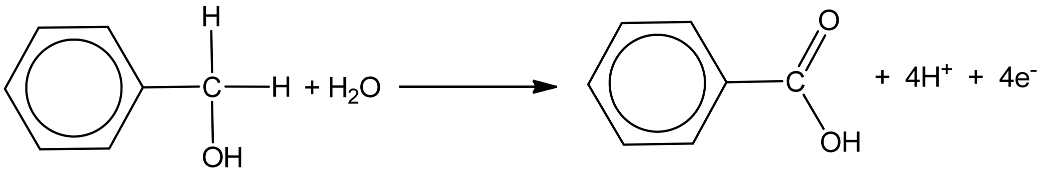
Bond angle: ....107^{o} **[accept anything btw 105^{o} to 109^{o}]**....

(iv) Suggest a reason why the reaction above would **not** occur if dimethylamine is replaced by trimethylamine, $(CH_3)_3N$.

There is <u>no H atom bonded to nitrogen</u> in $(CH_3)_3N$ and a removal of H bonded to nitrogen is required to form H_2O in the amide formation

(c) Benzoic acid can be produced by oxidising phenylmethanol, $C_6H_5CH_2OH$. A suitable oxidising agent for this reaction is hot acidified $KMnO_4$.

The half equation for the oxidation of phenylmethanol is as follows:



phenylmethanol benzoic acid

(i) State the colour change observed when phenylmethanol reacts with hot acidified $KMnO_4$.

Purple $KMnO_4$ decolourises/ turns colourless

(ii) Use the *Data Booklet* and the half equation given above to construct a balanced equation for the oxidation of phenylmethanol by hot acidified $KMnO_4$.

$4MnO_4^- + 5C_6H_5CH_2OH + 12H^+ \rightarrow 4Mn^{2+} + 5C_6H_5CO_2H + 11\ H_2O$

(iii) A pipette was used to transfer 25.0 cm^3 of 0.15 mol dm^{-3} of phenylmethanol into a conical flask. Calculate the minimum volume of 0.20 mol dm^{-3} of acidified $KMnO_4$ required to completely oxidise the phenylmethanol in the conical flask.

No. of moles of $C_6H_5CH_2OH$ = 25/1000 x 0.15 = **3.75 x 10^{-3}**

From (ii), 4 $KMnO_4$ ≡ 5 $C_6H_5CH_2OH$

No. of moles of $KMnO_4$= 4/5 x 3.75 x 10^{-3}

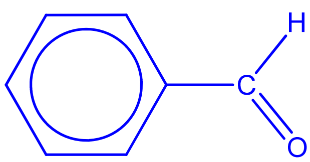
= **0.00300**

Min. vol. of $KMnO_4$= 0.00300 ÷ 0.2

=0.0150 dm^3 or 15.0 cm^3

[2]

(iv) Draw the structure of the product formed when phenylmethanol reacts with acidified $K_2Cr_2O_7$ with immediate distillation (controlled oxidation).



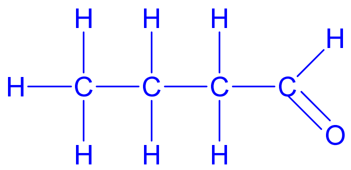
[Total: 15]

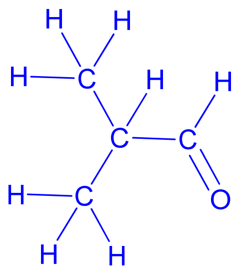
4 Carbon monoxide is a reagent used in an industrial process called hydroformylation. During hydroformylation, high pressure is used and the carbon monoxide reacts with an alkene and hydrogen gas to form an aldehyde.

The equation below shows an example of this process.

$$CO(g) + CH_3CH{=}CH_2(g) + H_2(g) \rightleftharpoons \underset{\text{aldehyde}}{C_4H_8O(g)}$$

(a) In the space below, draw the two possible constitutional isomers of C_4H_8O that are produced in the reaction above.





[2]

(b) The hydroformylation process is important because aldehydes are easily converted into many products. State the type of reaction that has occurred when an aldehyde is converted to a carboxylic acid.

Oxidation ....

(c) While the hydroformylation process uses rhodium catalyst in its soluble form, elemental rhodium is used in its nanoparticle form as a *heterogeneous catalyst* for other reactants. Explain the term *heterogeneous catalyst* as fully as you can.
Briefly explain why rhodium in its nanoparticle form works as a better catalyst than in its bulk form.

A catalyst is a substance that increases the rate of reaction by providing an **alternative reaction pathway with a lower energy barrier** (lower Ea) for the reaction to take place.

A heterogeneous catalyst is a catalyst that is in a **different phase** from the reactants.

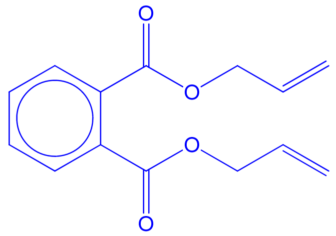
Rhodium in its nanoparticle form has a **large surface area** (or large surface to volume ratio) **for reaction to take place more quickly**.

....[3]

[Total: 6]

5 Poly(diallyl phthalate) is an example of a cross linked, thermoset polymer. It is suitable for use in high-performance military electrical components because of its ability to retain its properties when subjected to extreme environmental conditions of high temperature and high humidity.

(a) **(i)** Draw the structure of the monomer of poly(diallyl phthalate) and state the functional group that allows the monomer to become a cross-linked polymer.



Alkene

[2]

(ii) Explain, with reference to its structure and bonding, why it is able to retain its properties under the extreme environmental conditions stated above.

The polymer is **macromolecular with extensive covalent bonds** in the cross links that are **strong and hard to break. Hence the polymer is able to withstand high temperatures**. The polymer is largely **unable to form** extensive **hydrogen bonds with water molecules** and hence it is able to withstand high humidity without degrading.

....[2]

(b) Calcium alginate is also a cross-linked polymer but its properties vary greatly from poly(diallyl phthalate). Each calcium ion can attach to two alginate polymer strands to form cross-links. Calcium alginate is used in wound dressings. When the dressing needs to be changed, the wound can be rinsed with a solution that will break the cross-links. Suggest a suitable solution that can be used and briefly explain why the cross links are broken.

Sodium chloride solution. **[any group 1 metal salt solution]**

The sodium ions displace the calcium ions. As each **sodium ion is singly charged and unable to attach two polymer strands** , the cross-links begin to fall apart.

....[2]

[Total: 6]

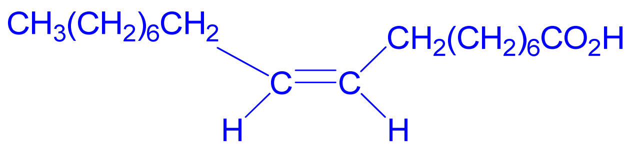
6 Oleic acid is a type of fatty acid that is present in avocados. Oleic acid has the following structure:

$$CH_3(CH_2)_7CH{=}CH(CH_2)_7CO_2H$$

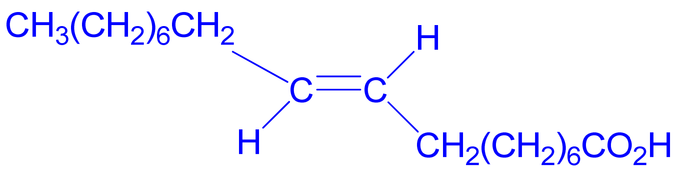
Oleic acid has the cis configuration around the C=C double bond, whereas elaidic acid is the trans isomer of oleic acid.

(a) Draw the condensed structural formulae of oleic acid and elaidic acid, showing clearly the configuration about the C=C in each acid.

oleic acid:



elaidic acid:



[2]

Avocado flesh turns brown rapidly upon exposure to oxygen in the air. In the presence of oxygen, enzymes in the avocado help in the formation of a class of compounds called quinones. Quinones are capable of producing polymers called polyphenols that give the brown colour.

(b)(i) Suggest a simple method that can be done at home to slow down the browning of avocado flesh. Give a brief explanation to support your suggestion.

Wrap it in cling wrap such that a barrier prevents oxygen from reacting with the enzymes. **Or** put the avocado in the fridge as the lowered temperature slows down the browning reaction / inhibits the enzyme from working as quickly. **Or** squeeze lemon / lime juice on the avocado as the lowered pH inhibits the enzyme from working as quickly. **[or any other *logical* suggestion that prevents oxygen from reaching the avocado flesh.] No credit given if the suggestion is correct but the reasoning is wrong.**

(ii) Define the term, *polymer*.

A polymer molecule is a macromolecule made up of monomers, with average molar mass of at least 1000 or at least 100 repeat units.

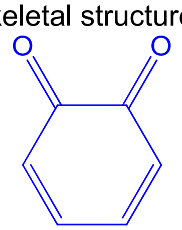
(iii) 1,2-benzoquinone is an example of a quinone. It contains 66.7% carbon and 3.7% hydrogen. The rest of the molecule comprises of oxygen which is present as two ketone functional groups in a molecule of 1,2-benzoquinone.
Calculate the empirical formula of 1,2-benzoquinone. Hence, deduce the molecular formula of 1,2-benzoquinone, stating clearly how you arrived at your answer.

	C	H	O
%	66.7	3.7	**29.6**
%/A_r	$\frac{66.7}{12.0}$=5.56	$\frac{3.7}{1.0}$= 3.70	$\frac{29.6}{16.0}$=1.85
Simplest ratio	3	2	1

Empirical formula is C_3H_2O
As each ketone functional group contains one oxygen atom and there are two ketone functional groups, the molecular formula is $C_6H_4O_2$. **[with valid explanation]**

[3]

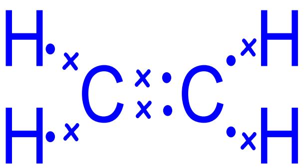
(iv) In a molecule of 1,2-benzoquinone, the two ketone functional groups are on adjacent carbon atoms in a cyclic structure. Using this information and your answer in **(iii)**, draw the skeletal structure of 1,2-benzoquinone.

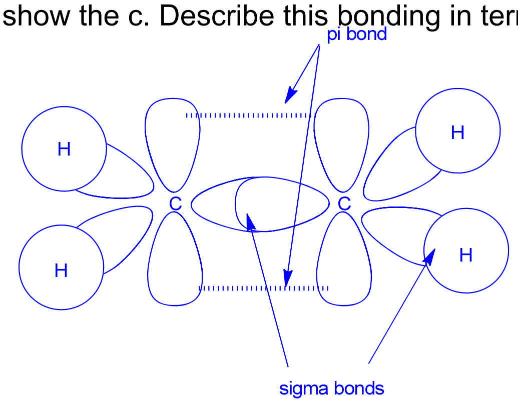


Other structures that are cyclic that satisfy the descriptions are accepted.

Avocados ripen faster if they are put in a plastic bag with some bananas. Bananas produce trace amounts of ethene gas which functions as a ripening hormone in fruits.

(c) Draw a dot-and-cross diagram to show the c. Describe this bonding in terms of orbital overlap.





The drawing is not necessary but clearly a labelled drawing may help in the descriptions.

Ethene has **5 sigma bonds and 1 pi bond**. **[or implied in description]**

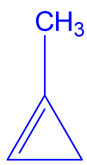
Sigma bonds are formed when an ***s* orbital of H head-on overlaps** with **an** (*sp*^{2}) **orbital of C** and when **an** (*sp*^{2}) **orbital of C head-on overlaps with an** (*sp*^{2}) **orbital of a neighbouring C.**

[4]

Pi bond is formed when the parallel ***p* orbital** of one C atom, **overlaps side-ways** with the *p* orbital of another C.

Bananas are usually picked from their plantations when they are unripe. But the production of ethene can quickly ripen the fruits during transportation. Hence, to slow down the ripening process, 1-methylcyclopropene is used because it binds tightly to the ethene receptor in the fruits and therefore blocks the effect of ethene.

(d) **(i)** Draw the structure of 1-methylcyclopropene.



(ii) Explain why 1-methylcyclopropene has a higher boiling point than ethene.

Both are simple molecules but **1-methylcyclopropene has more electrons** ....

hence it has **stronger instantaneous dipole – induced dipole forces of** .....

attraction between molecules that need more energy to overcome. ....

[2]

[Total: 15]

Section B

Answer **one** question in the spaces provided.

7(a) Carbonic acid is a weak acid that partially dissociates in water according to the following equation:

$$H_2CO_3(aq) + H_2O(l) \rightleftharpoons HCO_3^-(aq) + H_3O^+(aq)$$

The K_a of carbonic acid is 4.5×10^{-7} mol dm^{-3}.

The hydrogencarbonate ion, $HCO_3^-(aq)$, can also behave as weak acid in water:

$$HCO_3^-(aq) + H_2O(l) \rightleftharpoons CO_3^{2-}(aq) + H_3O^+(aq)$$

The K_a of hydrogencarbonate is 5.0×10^{-11} mol dm^{-3}.

(i) State the relationship between carbonic acid and the hydrogencarbonate ion.

Conjugate acid-base pair / hydrogencarbonate is the conjugate base of carbonic acid, and the difference between the two is only one proton.

(ii) Write the K_a expression for hydrogencarbonate ion and briefly explain why this K_a value is much smaller than the K_a value for carbonic acid.

$$K_a = \frac{[CO_3^{2-}][H_3O^+]}{[HCO_3^-]}$$

It is **more difficult for a positive H^+ to be lost from an negatively charged ion** HCO_3^- due to stronger electrostatic forces of attraction, than from a neutral molecule H_2CO_3. [2]

(iii) The hydrogencarbonate ion is part of the buffer in the blood.
Define *buffer* and explain how $HCO_3^-(aq)$ reacts with acid in the blood. Support your answer with a suitable equation.

A buffer is a solution that maintains a fairly constant pH when small amounts of acid or alkali are added to it.

When blood gets more acidic, the H^+ is removed by HCO_3^-, hence keeping the pH of blood almost constant. $HCO_3^- + H^+ \rightarrow H_2O + CO_2$ [2]

(b) The hydroxyl radical, represented by •OH, is highly reactive and short lived. It is often referred to as the "detergent" of the atmosphere because it reacts with greenhouse gases such as methane.

$$CH_4 + •OH \rightarrow •CH_3 + H_2O$$

A study of the rate of this reaction in a mixture, gave the following results. The initial concentration of methane for this study was much larger than concentration of hydroxyl radical and assumed to be constant throughout the reaction.

Time after start of reaction / x 10^{-4} s	**relative [•OH]**
0.0	10.0
1.0	7.0
2.0	5.0
3.0	3.5
4.0	2.5
5.0	1.8

(i) Define the term *half-life* of a reaction.
Use the data in the table to deduce the half-life of the reaction and to show that the reaction is first order with respect to [•OH].

Half-life is the time taken for the concentration of reactant to fall to half its original value.

The half-life of the reaction is 2.0 x 10^{-4} s.

Time taken for the relative [•OH] to fall from 10.0 to 5.0 and from 5.0 to 2.5 is constant at 2.0 x 10^{-4} s hence the reaction is first order with respect to [•OH].

[3]

(ii) Assume that the order of reaction with respect to [CH_4] is 1, state how the half-life of the reaction would change if:

Initial [CH_4] is doubled:

Half-life will be halved the original value (or half-life will be 1.0 x 10^{-4} s)

Initial relative [•OH] is halved:

Half-life will stay the same (or half-life will still be 2.0 x 10^{-4} s) [2]

(c) Describe reactions that illustrate the variation in acid-base behaviour of the oxides of the elements in Period 3, using MgO, Al_2O_3, and SO_3 as examples.
Write equations for all the reactions you describe.

Across Period 3, the oxides vary **from basic to amphoteric to acidic**.

MgO is a basic oxide which reacts with acid to give salt and water.

$MgO + 2HCl \rightarrow MgCl_2 + H_2O$

Al_2O_3 is an amphoteric oxide which reacts with both acids and alkalis.

$Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$

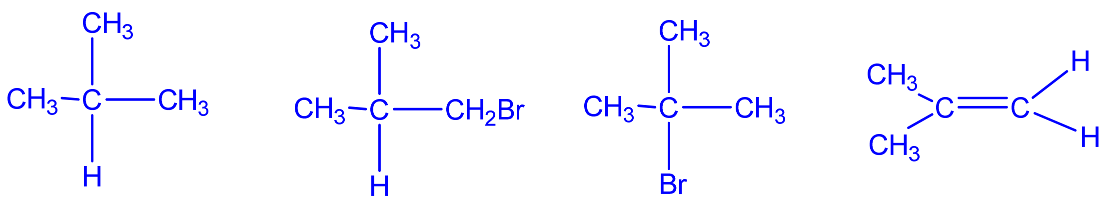
$Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2NaAl(OH)_4$

SO_3 is an acidic oxide which reacts with alkali to give salt and water.

$SO_3 + 2NaOH \rightarrow Na_2SO_4 + H_2O$ [5]

(d) Alkane **A**, C_4H_{10}, reacts with bromine under ultraviolet light to produce only two different mono-brominated products, **B** and **C** in the likely ratio of 9:1. When reacted with alcoholic KOH, both **B** and **C** produce hydrocarbon **D**, C_4H_8.

(i) Suggest structures for **A**, **B**, **C** and **D**. Label the structures clearly.



A: C_4H_{10} B C D: C_4H_8

[3]

(ii) Describe a simple chemical test to distinguish between **A** and **D**.

To separate test tubes containing aqueous bromine , bubble gases **A** and **D** into two different test tubes. Orange aqueous bromine decolourises in the presence of **D** but remains orange in the presence of **A**.

[2]

OR To separate test tubes containing bromine in CCl_4, bubble gases **A** and **D** into two different test tubes. Reddish-brown bromine decolourises in the presence of **D** but remains reddish-brown in the presence of **A**.

[Total: 20]

8(a) The table below gives the acid dissociation constants, K_a, of three acids at 25 °C.

Acid	**Formula**	K_a **/ mol dm^{-3}**
Methanoic acid	HCO_2H	1.8×10^{-4}
Ethanoic acid	CH_3CO_2H	1.5×10^{-5}
Chloroethanoic acid	CH_2ClCO_2H	1.3×10^{-3}

(i) From the K_a values, identify the strongest of the three acids and write the K_a expression for the acid you identified.

Chloroethanoic acid

$$K_a = \frac{[CH_2ClCO_2^-][H^+]}{[CH_2ClCO_2H]}$$

[2]

(ii) A student needed to select one of the three acids above to prepare a buffer solution of pH 3.80. The acid chosen has to have a pK_a that is closest to the pH value of the buffer solution.

Define the term *buffer*.
Identify the acid chosen for the preparation of the buffer and briefly explain how the buffer can be prepared (calculation of actual amounts is not required.)

A buffer is a solution that maintains a **fairly constant pH** when **small amounts** of acid or alkali are added to it.

Methanoic acid (pK_a is 3.74).

Add about an equal amount of **sodium methanoate or potassium methanoate to methanoic acid.** [3]

OR

Add **sodium hydroxide to a solution of methanoic acid ensuring that methanoic acid is in excess** (or sodium hydroxide is limiting.)

(b) 2-bromopropane, $CH_3CHBrCH_3$, can be hydrolysed by KOH(aq) to form propan-2-ol. Results of an investigation into the kinetics of this reaction are given below.

Expt number	$[CH_3CHBrCH_3]$ / mol dm^{-3}	[KOH] / mol dm^{-3}	Relative initial rate
1	0.10	0.20	1.00
2	0.60	0.40	6.00
3	0.30	0.20	3.00

(i) Define the term *order of reaction*.
Use the data in the table to deduce the order of reaction with respect to each reagent.

The order of reaction with respect to a given reactant is defined as the power to which the reactant concentration is raised to in the experimentally determined rate equation.

Comparing expts 1 and 3, when [KOH] is constant and $[CH_3CHBrCH_3]$ increases by 3 times, relative initial rate increases by 3 times. Hence reaction is first order with respect to $[CH_3CHBrCH_3]$.

Knowing that the reaction is first order with respect to $[CH_3CHBrCH_3]$, when comparing expts 1 and 2 where $[CH_3CHBrCH_3]$ increases by 6 times, rate increases by 6 times also. This indicates that the increase in [KOH] by 2 times has no impact on the rate. Hence reaction is zero order with respect to [KOH]. **[or correct mathematical method]**

[3]

(ii) Write an overall rate equation for the reaction between $CH_3CHBrCH_3$ and KOH and state the units of the rate constant.

rate = k $[CH_3CHBrCH_3]$ and units of k is s^{-1} (accept min^{-1}).

[2]

(c) The oxides Na_2O, Al_2O_3, and P_4O_{10} differ considerably in their chemical properties. Describe what happens when separate samples of each oxide are added to water Give equations where appropriate and state clearly why, if no reactions occur.

What is the effect of adding universal indicator to each resulting solution?

$Na_2O + H_2O \rightarrow 2NaOH$,

Na_2O **dissolves readily** in water to form an alkaline solution, which will turn universal indicator **purple/blue**.

Al_2O_3, has **does not dissolve** due to its **high lattice energy**, hence the universal indicator should **remain green.**

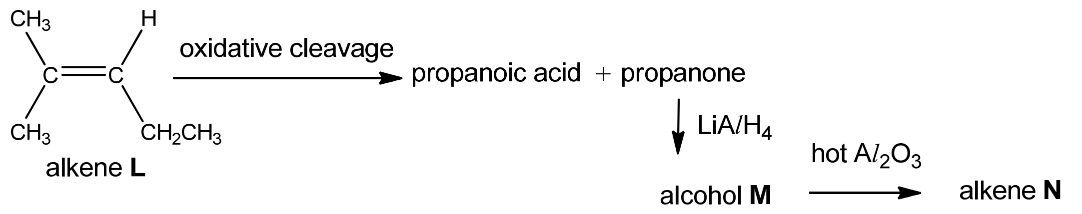
$P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$

P_4O_{10} **hydrolyses readily in water** to form an acidic solution, which will turn universal indicator **red/orange**.

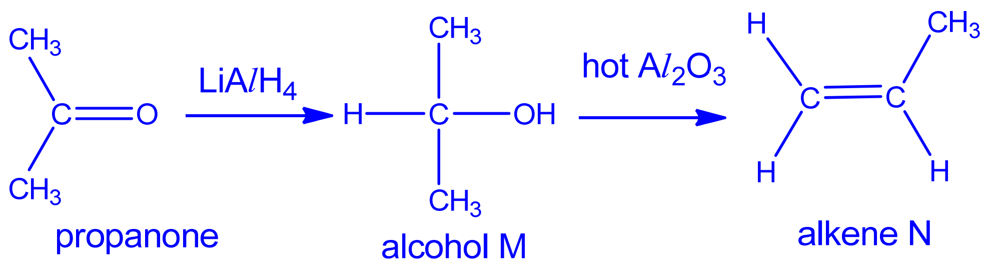
[5]

(d) When reacted with strong oxidising agents under high temperature, alkenes can undergo *oxidative cleavage* reactions. Such reactions involve the complete breaking of the C=C bonds in the alkenes to give carbon-oxygen bonds such as C-O or C=O. These new bonds are part of functional groups such as carboxylic acids and ketones.

Alkene **L** undergoes oxidative cleavage to produce propanoic acid and propanone. Propanone is able to undergo further reactions. The reaction scheme is shown below.



(i) Deduce the structures of **M** and **N**. [2]

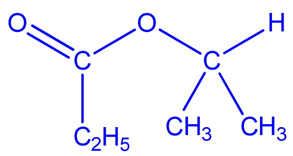


Propanoic acid produced from alkene **L** reacts with alcohol **M** to give a sweet smelling organic compound.

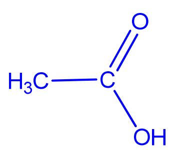
(ii) State the type of reaction that has taken place.

Condensation

(iii) Draw the structure of the organic product produced.



(iv) Draw the structure of the organic product produced when alkene **N** undergoes oxidative cleavage.



[Total: 20]